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SER402 Capstone Research

Terms:
BLE - Bluetooth Low Energy
Offset_ms - time difference between detection and notification
λ - value to be determined

Problem

**Accurate* incident GPS locations should be provided to the user. The IBDC device is responsible for capturing incidents as they occur and sending incident information over Bluetooth to our application. The inter-device communication over Bluetooth introduces three issues:

1. There is elapsed from when an event occurs to when the phone receives the event.
2. The phone application is responsible for capturing GPS location data.
3. There may be connection issues i.e. dis-connection when an event occurs.

These issues directly affect the accuracy of incident GPS locations. I would like to state some assumptions, define metrics and then produce an executable plan for solving the above issues in regard to accurate incident GPS locations.

Assumptions

- I am assuming that the time offset in milliseconds accounts for the creation of the event to transmission of the event from the IBDC device. The time the message travels over the BLE connection and the time it takes for the application to parse the message is not accounted for. This means (current_time - offset_ms) > true time of the incident.
- If the device provides an offset more than λ milliseconds, there has been a fatal error. A GPS location can not be estimated within an appropriate margin of error.
- The most accurate a GPS ping can be is within +-10 meters. This is the most accurate a ping can be, not a guarantee that each ping will be +-10 meters.
- Physical limitations of devices must be taken into account: storage; battery usage; data usage.

GPS Location Accuracy Requirements

GPS location data must be evaluated in terms of two independent measures: **temporal accuracy** and **spatial accuracy**.

1. Temporal Accuracy

Temporal accuracy describes the difference between the timestamp of a GPS measurement and the actual time an incident occurred.

The temporal error is defined as:

$E_t = |t_{GPS} - t_{incident}|$

Where:

- t_{GPS} represents the timestamp of the GPS measurement.
- $t_{incident}$ represents the timestamp of the detected incident.

Ideally, the GPS measurement would occur at the exact moment of the incident:

$E_t = 0$

However, due to GPS sampling intervals, communication latency, and system constraints, exact temporal alignment cannot be guaranteed.

Therefore, our objective is to minimize temporal error:

$\min E_t$

2. Spatial Accuracy

Spatial accuracy describes the geographic distance between the estimated incident location and its true location.

Let:

$P_{estimated} = (\hat{\varphi}, \hat{\lambda})$

Where:

- $\hat{\varphi}$ represents latitude.
- $\hat{\lambda}$ represents longitude.
- The hat notation indicates an estimated value.

Ideally, the estimated and actual coordinates would be identical:

$P_{estimated} = (\hat{\varphi}, \hat{\lambda}) = P_{actual} = (\varphi, \lambda)$ (unkown)

Therefore, our objective is to minimize the geographic distance between these positions:

$E_{\square} = d(P_{estimated}, P_{actual})$

Where (d) represents geographic distance measured in meters.

There are two factors of variability to be accounted for when using GPS location data in this scenario.

- Variability introduced by using GPS inherently, in our case with the chosen GPS location settings, at best +-10 meters.
- Variability introduced by taking the GPS ping at a distance away from the point of the accident.

Note: *We can only control for GPS ping at a distance away from the actual incident location. The variability introduced by GPS can be reduced, but it would suggest changes to our chosen design solution.*

3. Exploring Potential Solutions

Now that we have defined what makes a GPS location accurate, we try to control those variables.

The best effort we have at minimizing $E_{\square}$ is using the time offset we already have in place. There is nothing more we can do without suggesting changes to the architecture of our system.

As for minimizing $E_{\square}$, we can try and backtrack to the user's location based on the time offset from the event. To do this, we propose storing GPS ping location data periodically throughout the duration of a ride. A history of GPS pings can be utilized in a few ways to determine the original position. Depending on the sampling frequency of GPS pings, the method for determining the incident's best estimated position changes.

Method: Time indexed
GPS Ping Frequency: 1 ping / second

Algorithm Steps:

1. Use time_offset to backtrack to closest stored GPS ping
2. Select that GPS coordinate as the best candidate

Accuracy in terms of E□ = d(P_estimated, P_actual)?

Method: Time indexed + Dead Reckoning
GPS Ping Frequency: 1 ping / 5 seconds

Algorithm Steps:

1. Use time_offset to backtrack to the closest stored GPS ping and set that as the decision GPS coord.
2. If there's an exact GPS coordinate, select that GPS coordinate as the best candidate
3. Else, find the closest GPS coordinate. Calculate the estimated position using dead reckoning.

Accuracy in terms of E□ = d(P_estimated, P_actual)?

Method: Dead reckoning
GPS Ping Frequency: 1 ping / 10 seconds

Algorithm Steps:

1. Use time off_set to backtrack to the closest stored GPS ping and set that as the decision GPS coordinate.
2. Input velocity and estimate the position the user would have been in as the result.

Accuracy in terms of E□ = d(P_estimated, P_actual)?

Mobile Development Considerations

The above proposals can estimate the time of the incident. Ideally, we can store a GPS location at every instant and estimate, within the margin of error of using GPS, where a user is about.

GPS Sampling Frequency — Session Examples

The following table illustrates the number of GPS measurements stored during a ride session based on the configured GPS sampling interval.

GPS Sampling Interval	1-Hour Session	2-Hour Session
1 ping / second	3,600	7,200
1 ping / 5 seconds	720	1,440
1 ping / 10 seconds	360	720

Calculation:

Total GPS Measurements = Ride Duration (seconds) / GPS Sampling Interval (seconds)

Observations:

- A 1-second sampling interval provides the highest temporal resolution but generates the largest number of stored GPS measurements.
- A 5-second sampling interval reduces the number of measurements by 80% compared with 1-second sampling.
- A 10-second sampling interval reduces the number of measurements by 90% compared with 1-second sampling.

Note: *These calculations assume continuous GPS collection at the specified interval without missed measurements or interruptions.*

Summary

The price of requiring and storing a GPS ping is not without cost. A paper written by my teammate Chris Paquin can be found here: GPS Cost. His research indicates that GPS drain does not have a noticeable effect on battery life in his researched case, but many online anecdotes and small studies come to the opposite conclusion. This leads me to believe that Android and Apple incorporate optimizations into their GPS coordinate calculations that reduce the effect GPS has on the battery life of the phone. Even with these assumed optimizations, I'm not confident that the application battery drain rate won't become noticeable over long rides. I'm going to implement all three proposed solutions above. As more testing is done with the devices, testers can determine which solution works best.